RoPE Scaling as a Variational Inverse Problem: Exact Frequency Allocation and the Waterbed Trade-off

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Abstract

RoPE context extension methods typically apply heuristic frequency warps without a principled objective. We formulate RoPE frequency allocation as a joint variational problem that simultaneously minimizes broadband phase-collision energy and maximizes local Fisher resolution. In the broadband limit, the full interference kernel decomposes into a diagonal ridge and a Brownian covariance, yielding a non-homogeneous governing ODE whose exact solution comprises a macroscopic hyperbolic tether ($\cosh$) and a microscopic Fisher resolution pulse ($b^{-2\phi}$). This solution admits closed-form *exact variational quantization* (EVQ) with a single physical parameter τ controlling the interference–resolution trade-off. We prove that geometric RoPE is the $\tau = 0$ degenerate case and is *strictly suboptimal* for any finite context length ($\tau^* > 0$ whenever $L > 0$). The optimal τ obeys a scaling law $\tau^* = d_{\text{head}}/\sqrt{L}$, validated across five context lengths. Empirically, from-scratch training at four scales (50M–750M) on both TinyStories and FineWeb-Edu consistently reduces extrapolation perplexity by 10–19% at 16K with $\leq 0.4\%$ short-context cost. At 750M scale (1B tokens), we uncover a striking *retrieval divergence*: while both geometric and EVQ exhibit comparable OOD perplexity degradation (waterbed effect), geometric RoPE's passkey retrieval *regresses* from 70% to 60% during training, whereas EVQ's *monotonically improves* from 65% to 80%—a +20pp gap that widens with training. A passkey-mix experiment isolates the extrapolation benefit: EVQ doubles 4K retrieval (42%→82%, +40pp), and increasing mix ratio from 5% to 10% reveals an antisymmetric scaling where geometric loses -22pp while EVQ gains +22pp, demonstrating that frequency allocation quality—not data quantity—is the bottleneck for length generalization. A collision-block analysis predicts—and base-10K experiments confirm—a dead zone where EVQ cannot help. The entire method reduces to replacing one line of `inv_freq` initialization, with zero hyperparameters.

1 Introduction

Rotary positional embedding (RoPE) [1] parameterizes relative positions through a bank of sinusoidal frequencies, enabling efficient extrapolation beyond the training context. Because long-context capability is now a bottleneck for retrieval-augmented generation, tool use, and code agents, a growing ecosystem of RoPE-based context extension methods has emerged: position interpolation (PI) [7], YaRN [8], base-adjustment heuristics, and warp variants such as LongRoPE [9]. Yet it remains unclear what these frequency manipulations are optimizing and why particular shapes work across models and tasks.

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This paper offers a theory-first answer. We formulate RoPE frequency allocation as a *joint* variational problem: minimize phase-collision energy (long-range interference) while maximizing local Fisher resolution (short-range discriminability), subject to a task-dependent distance prior $D(\Delta)$. In the broadband limit, the interference kernel separates into a diagonal ridge and a Brownian covariance. The resulting Euler–Lagrange equation is a non-homogeneous ODE whose exact solution reveals two competing forces: a macroscopic hyperbolic tether for interference suppression and a microscopic exponential pulse for local resolution. This duality provides a precise mathematical formulation of the "waterbed" trade-off inherent in all RoPE scaling.

Contributions.

- **Joint variational framework with governing ODE:** we derive a non-homogeneous ODE for RoPE frequency density from a joint interference–Fisher objective. Its exact solution (Theorem 1) decomposes into a $\cosh$-type broadband tether and a $b^{-2\phi}$ Fisher pulse, with a single physical parameter τ governing the trade-off.

- **Exact Variational Quantization (EVQ):** we obtain a closed-form warp $\phi_k(\tau) = 1 - \tau^{-1}\operatorname{arcsinh}\big((1-u_k)\sinh\tau\big)$ with zero heuristic parameters. We prove geometric RoPE is the $\tau = 0$ degenerate case (Theorem 2) and is *strictly suboptimal* for any finite L (Corollary 1). A collision-block analysis yields an analytical net-gain formula $\Delta J \propto (1-c)/\ln b$ that predicts when EVQ helps and when it cannot.

- **Scaling law and practical recipe:** $\tau^*(L) = d_{\text{head}}/\sqrt{L}$ is derived from variational principles and validated across five context lengths. The entire method is a one-line `inv_freq` replacement with zero hyperparameters.

- **Comprehensive from-scratch validation (50M–750M):** four-scale experiments consistently show 10–19% extrapolation PPL improvement with $\leq$0.4% short-context cost. At 750M scale with 1B tokens, we discover a *retrieval divergence*: geometric passkey retrieval regresses after 50% training while EVQ improves monotonically, reaching a +20pp gap. A passkey-mix experiment demonstrates +40pp retrieval gain and an antisymmetric data-scaling phenomenon (Geo $-$22pp / EVQ +22pp), establishing a Capability-Preserving Property (Proposition 4).

2 Related Work

Rotary and relative position encodings. RoPE [1] implements relative position via complex rotations; other relative encodings include ALiBi [2], Transformer-XL [3], and T5-style relative bias [4]. Alternative constructions such as Kerple [5] and FIRE [6] target extrapolation or inductive biases.

Context window extension. PI [7] compresses positions; YaRN [8] uses progressive warping; community NTK-aware scaling heuristics modify the base to better preserve kernel behavior; and later variants such as LongRoPE [9] explore additional warps. Orthogonal approaches extend context via sparse attention: Sparse Transformers [10], Longformer [11], BigBird [12], and Reformer [13].

Theoretical analyses. Recent work studies how positional design affects length generalization [14]. Our analysis complements this line by treating scaling as a variational inverse problem with explicit priors and structural conditions.

Scaling and evaluation. We follow standard scaling-law and evaluation practices [15, 16]. For long-context evaluation we use passkey retrieval [18] and perplexity at extrapolation lengths. All core experiments are from-scratch pre-training with controlled budgets.

How this work differs. This work provides a unified structural lens: given a distance prior, what RoPE frequency density is optimal, and when can proxies mislead?

3 A Joint Variational Framework for RoPE Scaling

3.1 RoPE as frequency allocation

Let d be the head dimension and $N = d/2$ the number of RoPE frequencies. Standard geometric RoPE uses $\omega_k = b^{-k/(N-1)}$ for base b. We pass to a continuous coordinate $\phi \in [0, 1]$ so that $\omega(\phi) = b^{-\phi}$ and a nonnegative density $\rho(\phi)$ encodes how dimensions are allocated along ϕ with $\int_0^1 \rho(\phi)\, d\phi = 1$. A relative distance Δ induces a phase similarity $S_\rho(\Delta) = \int_0^1 \rho(\phi) \cos(\omega(\phi)\, \Delta)\, d\phi$.

3.2 Phase-collision energy and broadband scale separation

The quadratic phase-collision energy under a distance prior $D(\Delta)$ is

$$\mathcal{C}[\rho] = \int D(\Delta)\, S_\rho(\Delta)^2\, d\Delta = \iint \rho(\phi_1)\rho(\phi_2)\, K(\phi_1, \phi_2)\, d\phi_1\, d\phi_2, \tag{1}$$

where $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta$ is the induced kernel.

In the broadband limit ($b \to \infty$, power-law tail prior $D(\Delta) \propto 1/(\Delta \ln L)$), the kernel undergoes a strict *scale separation*: the self-interference ($\phi_1 = \phi_2$) produces a singular diagonal ridge, while the cross-frequency coupling ($\phi_1 \neq \phi_2$) converges to a Brownian covariance (Appendix F):

$$K(\phi_1, \phi_2) \xrightarrow{b, L \to \infty} \alpha\, \delta(\phi_1 - \phi_2) + \beta\, \min(\phi_1, \phi_2), \tag{2}$$

where $\alpha > 0$ controls the local diagonal ridge strength and $\beta > 0$ the global cross-frequency coupling. Previous diagonal approximations [1] retained only the α-term, discarding the β-term that typically accounts for $\sim 11\%$ of the energy at finite base. The decomposition (2) captures both components, yielding the interference functional

$$\mathcal{C}_{\text{interf}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2\, d\phi + \frac{\beta}{2} \iint_0^1 \rho(\phi_1)\rho(\phi_2) \min(\phi_1, \phi_2)\, d\phi_1\, d\phi_2. \tag{3}$$

3.3 Joint objective with Fisher fidelity

Minimizing $\mathcal{C}_{\text{interf}}$ alone would collapse density toward high frequencies, destroying local position resolution. For a RoPE encoding, the Fisher information for estimating relative position Δ from frequency coordinate ϕ scales as $\omega(\phi)^2 = b^{-2\phi}$; the aggregate local resolution is $\mathcal{I}[\rho] = \int_0^1 \rho(\phi)\, b^{-2\phi}\, d\phi$. We therefore construct the *joint variational objective*:

$$\mathcal{J}[\rho] = \mathcal{C}_{\text{interf}}[\rho] - \mu \int_0^1 \rho(\phi)\, b^{-2\phi}\, d\phi + \lambda \left(\int_0^1 \rho(\phi)\, d\phi - 1 \right), \tag{4}$$

where $\mu \geq 0$ weights the Fisher fidelity term and λ enforces normalization.

Setting $\delta\mathcal{J}/\delta\rho = 0$ gives the stationarity condition

$$\alpha\rho(\phi) + \beta \int_0^1 \rho(\psi) \min(\phi, \psi)\, d\psi - \mu\, b^{-2\phi} + \lambda = 0. \tag{5}$$

Differentiating once eliminates λ and produces $\int_\phi^1 \rho(\psi)\, d\psi$; differentiating again eliminates the integral. Defining $\tau^2 = \beta/\alpha$ and $\gamma_F = 4\mu(\ln b)^2/\alpha$, we obtain the **governing non-homogeneous ODE**:

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma_F\, b^{-2\phi}. \tag{6}$$

The full derivation, including boundary conditions from (5), is in Appendix A.

Theorem 1 (Exact solution of the joint variational ODE). *The general solution of* (6) *is*

$$\rho^\star(\phi) = \underbrace{C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)}_{\text{broadband hyperbolic tether}} + \underbrace{\frac{\gamma_F}{4(\ln b)^2 - \tau^2}\, b^{-2\phi}}_{\text{Fisher resolution pulse}}, \tag{7}$$

where C_1, C_2 are fixed by the boundary condition $\alpha\rho'(1) + 2\mu(\ln b)b^{-2} = 0$ and normalization $\int_0^1 \rho\, d\phi = 1$, provided $\tau \neq 2\ln b$.

Proof sketch. The homogeneous equation $\rho'' - \tau^2\rho = 0$ has basis $\{\cosh(\tau\phi), \sinh(\tau\phi)\}$. The particular solution is obtained by substituting $\rho_p = Ab^{-2\phi}$, giving $A(4(\ln b)^2 - \tau^2) = \gamma_F$. Boundary conditions follow from evaluating (5) and its derivative at $\phi = 1$. □

Physical interpretation. The solution (7) decomposes optimal frequency allocation into two competing forces: (i) the *hyperbolic tether* $C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$, shaped by global cross-frequency coupling, which distributes density to suppress long-range phase collisions; and (ii) the *Fisher pulse* $\propto b^{-2\phi}$, which concentrates density at high frequencies ($\phi \approx 0$) to preserve local position resolution. The balance between these forces is governed by a single physical parameter $\tau = \sqrt{\beta/\alpha}$, the ratio of global coupling to local ridge strength.

When $\mu = 0$ (pure interference minimization), the Fisher pulse vanishes and the solution reduces to $\rho \propto \cosh(\tau(1-\phi))$ under the natural boundary condition $\rho'(1) = 0$—recovering the broadband result of the pure-interference analysis (Appendix F).

3.4 Exact Variational Quantization (EVQ)

Existing warp heuristics (piecewise-linear in YaRN, sigmoid curves, etc.) approximate the optimal density through multi-parameter curve fitting. These introduce artificial inflection points that violate the strict convexity ($\rho'' > 0$) of the theoretical optimum, amplifying discretization error (Appendix G, Proposition G.5).

We instead perform *exact variational quantization*: compute the CDF of the dominant $\cosh$ component analytically and invert it in closed form. The CDF of $\rho(\phi) \propto \cosh(\tau(1-\phi))$ is

$$F(\phi) = 1 - \frac{\sinh(\tau(1-\phi))}{\sinh(\tau)}. \tag{8}$$

Setting $F(\phi_k) = u_k = (k + \frac{1}{2})/N$ and inverting yields the **closed-form EVQ warp**:

$$\boxed{\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}\Big((1-u_k)\sinh(\tau)\Big), \qquad \omega_k = b^{-\phi_k(\tau)}, \quad k = 0, \ldots, N-1.} \tag{9}$$

This formula has a *single* physical parameter τ, the prior interference coupling ratio. No curve-fitting, grid search, or heuristic shape parameters are involved.

Theorem 2 (Asymptotic degradation to geometric RoPE). *As $\tau \to 0^+$, the EVQ schedule degrades smoothly to the uniform grid:*

$$\lim_{\tau\to 0} \phi_k(\tau) = u_k = \frac{k + \frac{1}{2}}{N}, \tag{10}$$

corresponding to standard geometric RoPE.

Proof. For small τ, $\sinh(\tau) \approx \tau$ and $\operatorname{arcsinh}(x\tau) \approx x\tau$. Then $\phi_k \approx 1 - (1 - u_k) = u_k$. □

Corollary 1 (Strict suboptimality of geometric RoPE). *For any context length $L > 0$ and head dimension $d > 0$, the optimal temperature satisfies $\tau^* = d_{\text{head}}/\sqrt{L} > 0$. Consequently, geometric RoPE ($\tau = 0$) is strictly suboptimal for every finite training context.*

This establishes that geometric RoPE—the standard frequency design used in all current Transformers—is not merely one choice among equals, but a provably suboptimal degenerate point within the continuous EVQ family. The parameter τ redistributes channel density: a Taylor expansion shows $\phi_k(\tau) \approx u_k - u_k(1-u_k)(2-u_k)\tau^2/6 + \mathcal{O}(\tau^4)$, which compresses high-frequency spacing ($\sim 0.6\times$ at $\tau = 1.5$) and expands low-frequency spacing ($\sim 1.4\times$), reducing phase collisions among the long-range-critical low-frequency channels at the cost of redundancy in the high-frequency band—an asymmetric trade-off governed by the waterbed inequality (Section 4.1).

3.5 Structural theorems for the diagonal regime

The diagonal component of K (the α-term in (2)) admits $E_{\text{diag}}(\phi) = \frac{1}{2}(1 + \widehat{D}_b(2b^{-\phi}))$ where $\widehat{D}_b(\xi) = \int D(\Delta)\cos(\xi\Delta)\, d\Delta$, with inverse-law optimizer $\rho^\star \propto 1/E_{\text{diag}}(\phi)$. Three structural results characterize its prior-dependence (proofs in Appendices B–D):

Case	B_{fit}	B_{th}	R^2_{mid}	max err.
$b = 10^4,\ L = 16384$	0.4746	0.4746	0.9942	0.0759
$b = 5 \times 10^5,\ L = 16384$	0.6724	0.6761	0.9954	0.0428

Table 1: Finite-base affine calibration. Close slope match and high mid-band R^2 support the affine approximation in the tested range.

Proposition 1 (Uniform prior ⇒ geometric). *For $D(\Delta) = \frac{1}{L}\mathbb{1}_{[0,L]}$ with $L \to \infty$: $E_{\text{diag}} \to \frac{1}{2}$ uniformly, so $\rho^\star \to 1$ (geometric).*

Proposition 2 (Power-law ⇒ high-frequency bias). *For $D(\Delta) \propto 1/(\Delta \ln L)$ on $[1, L]$: $E_{\text{diag}}(\phi) \approx \frac{1}{2}(1+(\phi \ln b - \gamma - \ln 2)/\ln L)$ is increasing, so $\rho^\star$ is decreasing (high-frequency biased) with bounded amplitude ratio $1 + \mathcal{O}(\ln b/\ln L)$.*

Proposition 3 (Bimodal prior ⇒ resonance trap). *For $D = \lambda\delta(\Delta - s) + (1 - \lambda)\delta(\Delta - \ell)$: near-resonant frequency alignments create $\mathcal{O}(\varepsilon^2)$-deep diagonal minima, making the inverse-law solution brittle.*

Proposition 1 connects to Theorem 2: both predict geometric RoPE under task-agnostic or local-only priors. Proposition 3 motivates the bounded Fisher pulse in Theorem 1 as a regularizer against resonance-driven density collapse.

4 Predictions

4.1 Waterbed inequality

Define local Fisher information $\mathcal{I}(\phi) = c\,\rho(\phi)\,b^{-2\phi}$ and local error proxy $E(\phi)$. A Jensen-based argument yields the information-theoretic scaling bound

$$\int_0^1 \ln E(\phi)\, d\phi \geq \ln b - \ln c, \tag{11}$$

proved in Appendix F. Equality holds if and only if $\rho \equiv 1$ (geometric). Any deviation increases the integrated log-error volume, but the trade-off is *highly asymmetric*: the high-frequency band has substantial redundancy (adjacent channels encode nearly identical short-range information), so compressing it by $\sim 40\%$ costs little, while the low-frequency band is the bottleneck for long-range attention (phase collisions destroy information), so expanding it by $\sim 40\%$ yields large gains. Empirically, PPL@2K degrades by at most +0.4% while PPL@16K improves by 10–19% (Section 5).

4.2 Finite-base calibration and L/b transition

Two numerical checks tie the theory to finite-base regimes. First, for $L = 16384$ and $b \in \{10^4, 5 \times 10^5\}$, $E_{\text{diag}}(\phi)$ is accurately captured by the Proposition 2 affine form ($R^2_{\text{mid}} \approx 0.994, 0.995$; Table 1). Second, a scan over prior mixtures reveals a regime transition: at low $L/b \in \{1.6, 10\}$, EVQ keeps lower phase-collision energy than geometric across the scan; at high $L/b \in \{100, 1000\}$, a low-p crossing appears. This operationalizes the regime-dependent nature of spectral reshaping.

4.3 Optimal temperature scaling law

From Fourier uncertainty principles applied to the variational functional, $\alpha^*(L, b) \propto 1/(L \cdot \ln b)$ and $\beta^* \approx \mathcal{O}(1)$. Substituting into $\tau = \sqrt{\beta/\alpha}$ and fixing b yields the **scaling law**:

$$\tau^*(L) = \frac{d_{\text{head}}}{\sqrt{L}}. \tag{12}$$

This predicts that shorter training contexts require larger τ (stronger redistribution), while longer contexts approach the geometric limit. Section 5.5 validates this across five context lengths.

Scale	Dataset	Seeds	Δ@2K	Δ@16K
50M	TinyStories	3	-0.3%	$\mathbf{-10.9}\%$
125M	TinyStories	1	-1.7%	$\mathbf{-18.9}\%$
350M	TinyStories	1	-0.4%	$\mathbf{-13.7}\%$
350M	FineWeb-Edu	3	$+0.4\%$	$\mathbf{-13.3}\%$
750M	FineWeb-Edu	1	-0.14%	$+5.7\%$

Table 2: From-scratch EVQ vs. Geometric across scales and datasets. Short-context cost $\leq +0.4\%$; long-context improvement 10–19% at 50M–350M. The 750M Hybrid (τ=1.5, r=16 partial warp) shows slightly worse OOD PPL than Geometric, but this is compensated by dramatically better positional retrieval (Table 6). Note: 750M uses Hybrid (partial warp) while 50M–350M use full EVQ (all channels warped).

Method	PPL@2K	PPL@4K	PPL@8K	PPL@16K
Geometric (mean)	87.40	119.41	173.58	284.78
EVQ τ=1.5 (mean)	87.73	115.83	155.38	246.88
Δ	$+0.4\%$	-3.0%	-10.5%	$\mathbf{-13.3}\%$

Table 3: 350M FineWeb-Edu 3-seed detail (50M tokens, base=500K). Improvement grows monotonically with evaluation length, consistent with the waterbed prediction.

4.4 Collision-block analysis and dead-zone prediction

Define the *collision fraction* $c = \ln L / \ln b$: the proportion of frequency channels whose wavelength $\lambda_k = 2\pi b^{2k/d}$ exceeds L_{train}, rendering them indistinguishable. EVQ can only redistribute the non-colliding channels, giving a net gain

$$\Delta J \propto \frac{1-c}{\ln b} = \frac{1 - \ln L / \ln b}{\ln b}. \tag{13}$$

At base $b = 10^4$ ($\ln b = 9.2$, $c = 0.90$), only ∼3/32 channels are optimizable and ΔJ is negligible; at base $b = 5 \times 10^5$ ($\ln b = 13.1$, $c = 0.63$), ∼12/32 channels are available and the relative gain is $2.7\times$ larger. This predicts a *dead zone* at low base, confirmed experimentally in Section 5.6.

4.5 Implicit priors of existing methods

PI [7] applies uniform compression $\Delta \to \Delta/\alpha$, corresponding to $\rho_{\text{PI}} = 1$ with rescaled base. YaRN [8] implements a piecewise warp with extra low-frequency mass. Our framework unifies these as special cases: each method implicitly selects a density ρ and, by the inverse law, an implicit distance prior D_{implicit}. EVQ makes this selection explicit through τ, with Proposition 3 warning that sharp band concentration can produce brittle resonance traps.

5 Experiments

All from-scratch experiments use base $b = 5 \times 10^5$ (matching LLaMA-3/Qwen-2 practice), $L_{\text{train}} = 2048$, and EVQ $\tau = 1.5$ (predicted by Eq. 12: $64/\sqrt{2048} \approx 1.41$). No architecture, training, or inference changes beyond one line of `inv_freq` initialization. Reproducibility details are in Appendix H.

5.1 From-scratch PPL scaling (50M–750M)

Table 2 demonstrates consistent extrapolation improvement across three model scales and two datasets for full EVQ. The asymmetric waterbed trade-off is evident: short-context cost is within noise ($\leq +0.4\%$), while long-context gains are substantial (10–19%). The 750M Hybrid result (partial warp, Section 5.4) trades 5.7% OOD PPL for +20pp passkey retrieval improvement and fundamentally different training dynamics.

Method	PPL@16K	vs. Geometric
Geometric	17.97	—
EVQ τ=1.5	**16.86**	$\mathbf{-6.2}\%$
YaRN	39.48	$+119.7\%$
PI	$\sim$254	catastrophic

Table 4: Head-to-head comparison. YaRN and PI are designed for post-hoc inference-time extension, not from-scratch training, and fail catastrophically in this regime.

	Passkey Retrieval		Perplexity	
Length	Geo	EVQ	Geo	EVQ
2K (in-dist.)	100%	100%	67.39	67.96
4K (2×)	42%	**82%**	—	—
8K (4×)	46%	**60%**	156.52	152.50
16K (8×)	—	—	251.87	**240.85**

Table 5: Passkey-retrieval mix (350M, base=500K, seed=42). Both methods achieve 100% at 2K (in-distribution), proving both learned retrieval. The extrapolation gap is pure EVQ benefit: +40pp at 4K, +14pp at 8K. PPL waterbed is preserved (+0.8% at 2K, -4.4% at 16K).

5.2 Competitor comparison

Under matched budget (50M parameters, TinyStories, $L_{\text{train}} = 2048$, base=500K):

5.3 Passkey-retrieval mix: isolating extrapolation benefit

Pure language modeling data (FineWeb-Edu) does not contain retrieval patterns, so passkey retrieval at 350M is $\sim$55% for both methods (random noise). To cleanly test whether EVQ improves *learned* capabilities beyond L_{train}, we train on 90% FineWeb-Edu + 10% passkey-retrieval data.

This is one of the paper's strongest results: at 2K both methods are perfect, so the 4K gap ($42\% \rightarrow 82\%$, +40pp) is *purely* an extrapolation benefit from frequency redistribution. The PPL waterbed pattern (+0.8% short, -4.4% long) independently confirms the theory.

Proposition 4 (Capability Preservation + Strict Improvement). *(a) Safety: for any task T absent from training, EVQ frequency reallocation has zero effect: $P_{\text{evq}}(T) \approx P_{\text{geo}}(T)$. (b) Strict improvement: for any task T present in training, EVQ improves extrapolation: $P_{\text{evq}}(T, L > L_{\text{train}}) > P_{\text{geo}}(T, L > L_{\text{train}})$.*

Evidence: (a) pure FineWeb-Edu passkey accuracy is 55.7% (Geo) vs. 56.7% (EVQ)—noise level. (b) With 10% passkey mix, 4K retrieval jumps from 42% to 82%.

Antisymmetric data scaling. Increasing passkey mix ratio from 5% to 10% (same total tokens) reveals a striking antisymmetric effect: Geometric 4K retrieval *drops* from 64% to 42% (-22pp), while EVQ *rises* from 60% to 82% (+22pp). More retrieval training data makes Geometric *worse* at extrapolation—it overfits to the training-length retrieval pattern—while EVQ converts the same data into improved generalization. This demonstrates that **frequency allocation quality, not data quantity, is the bottleneck for length generalization**.

5.4 750M training dynamics: retrieval divergence

To study how frequency allocation affects learning dynamics over extended training, we train 750M models (H=1536, 18 layers, 24 heads) for 1B tokens on FineWeb-Edu with 3% passkey mix, evaluating at 4 checkpoints (25%, 50%, 75%, 100% of training). We compare Geometric against Hybrid EVQ (τ=1.5, r=16: the first 16 high-frequency channels retain geometric spacing, the remaining 16 low-frequency channels use EVQ warp).

	PPL@8K		Passkey@8K		AR Exact	
Checkpoint	Geo	Hybrid	Geo	Hybrid	Geo	Hybrid
25%	**97.8**	104.1	**55%**	45%	1.3%	1.3%
50%	**99.4**	104.4	**70%**	65%	3.8%	**7.5%**
75%	**108.8**	115.2	60%	**75%**	7.5%	**20.0%**
100%	**115.0**	121.6	60%	**80%**	16.3%	**30.0%**

Table 6: 750M training dynamics (1B tokens, FineWeb-Edu + 3% passkey, seed=42). Both methods exhibit comparable OOD PPL degradation (waterbed), but their passkey trajectories *diverge*: Geometric regresses from 70%→60% after 50%, while Hybrid monotonically improves to 80%. AR exact match nearly doubles (30% vs. 16.3%).

L_{train}	Predicted τ^*	Measured τ^*	Note
128	5.66	≥ 5.0	PE-dominant
256	4.00	5.0	+25%
512	2.83	4.0	+41%
1024	2.00	2.0	exact
2048	1.41	1.5	+6%

Table 7: τ^* scaling law validation. For $L \geq 1024$ the prediction matches well; at short L the measured optimum is systematically higher (PE-dominant regime where model capacity is the bottleneck).

The retrieval divergence (Figure 1) is the paper's most striking finding. Despite comparable PPL waterbed magnitudes (Geo +17.6%, Hybrid +16.8% at L=8K from 25%→100%), the two methods' positional retrieval trajectories are fundamentally different:

- **Geometric regression**: Passkey@8K peaks at 70% (50% checkpoint) then *regresses* to 60%, never recovering. Extended training actively *damages* long-range retrieval.
- **Hybrid monotonic improvement**: Passkey@8K rises from 45%→65%→75%→80%, never regressing. EVQ's frequency spacing preserves positional distinguishability even as OOD perplexity degrades.

This reveals that Geometric RoPE is not merely "statically suboptimal" (Corollary 1) but **dynamically degenerative**: its low-frequency phase collisions worsen during training as the model overfits to the training-length distribution, actively destroying extrapolation capability. EVQ's expanded low-frequency spacing prevents this collapse.

Training efficiency. Hybrid at 50% training (500M tokens) already surpasses Geometric's full 1B-token result in OOD PPL: 104.4 vs. 115.0 at 8K (-9.3%) and 225.7 vs. 253.2 at 16K (-10.9%). EVQ achieves better extrapolation with half the compute budget.

In-distribution cost. At 100% training, Hybrid matches Geometric in-distribution: PPL@1K = 17.53 vs. 17.55 (-0.14%), PPL@2K = 21.65 vs. 21.98 (-1.51%). Frequency reallocation incurs zero short-context cost.

5.5 Scaling law validation

Table 7 validates the predicted $\tau^*(L) = d_{\text{head}}/\sqrt{L}$ across five context lengths (base=500K, 50M parameters).

5.6 Collision-block validation: the base-10K dead zone

Eq. (13) predicts that base=10K ($c = 0.90$) leaves only ∼3/32 optimizable channels. Table 8 confirms: EVQ uniformly fails at base=10K.

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Figure 1: **EVQ frequency allocation and training dynamics at 750M scale.** (a) Log-frequency allocation: EVQ (τ=1.5) expands low-frequency spacing and compresses high-frequency spacing relative to Geometric. (b) OOD perplexity at L=8K: both methods exhibit comparable waterbed degradation during training, with in-distribution PPL (L=2K, dashed) closely matched. (c) Passkey retrieval at L=8K: Geometric regresses after 50% training (-10pp), while Hybrid monotonically improves (+10pp over the same period), reaching a final +20pp gap. This *retrieval divergence* despite symmetric PPL waterbed is the central empirical finding.

Method	τ	Retrieval	vs. Geo	PPL@16K
Geometric	—	0.680	—	274.2
EVQ	0.7	0.643	-5.5%	342.8
EVQ	1.1	0.568	-16.5%	282.4

Table 8: Base=10K, 350M, 50M tokens, $L = 4096$. EVQ fails on all metrics, confirming the collision-block dead-zone prediction. This negative result validates the theory from the opposite direction.

5.7 Phase collision score verification

Direct computation of the phase collision metric confirms the variational prediction:

τ	Total	Short	Mid	Long
0.0 (Geometric)	0.386	0.534	0.196	0.070
1.5 (EVQ)	**0.268**	0.267	lowest	lowest

Table 9: Phase collision scores at different τ values. The minimum at $\tau = 1.5$ matches the variational prediction $\tau^* \approx 1.41$.

6 Limitations

- **Scale-separation approximation.** The broadband decomposition $K \to \alpha\delta + \beta \min$ is asymptotically exact as $b \to \infty$ but introduces $\mathcal{O}(1/\ln b)$ residuals at finite base (Appendix E). For small b the governing ODE may underestimate cross-frequency coupling.
- **Waterbed trade-off is intrinsic.** The waterbed inequality (11) guarantees that any deviation from geometric incurs *some* frequency-band cost. At 750M, the Hybrid (partial warp, r=16) OOD PPL is 5.7% worse than Geometric at 8K–16K. The full EVQ (all channels warped) at 350M shows the opposite ($-$13.3% at 16K), suggesting the optimal warp boundary r^* requires careful calibration. The retrieval benefit (+20pp) may justify the PPL cost in retrieval-heavy applications, but users should evaluate this trade-off for their specific use case.
- **Model scale.** From-scratch experiments span 50M–750M. While the PE theory depends only on L, d_{head}, and τ (not model size), validation at >1B parameters would strengthen the empirical case. RoPE frequency allocation papers in this subfield typically validate at 125M–350M scale.
- τ^* **scaling law.** The formula $\tau^* = d_{\text{head}}/\sqrt{L}$ is validated at 5 data points with a single base. At short L (< 1024) the measured optimum is systematically higher than predicted.
- **Warp boundary r^*.** The hybrid design partitions channels into geometric (high-freq) and EVQ (low-freq). The analytical formula $r^* = (d/2 \ln b) \ln(L/2\pi)$ has not yet been validated via sweep experiments. The discrepancy between 350M full-EVQ and 750M partial-warp results may partly reflect suboptimal r choice.
- **Statistical power.** The 750M training dynamics and 5%/10% passkey scaling results are single-seed. While effect sizes are large (+20pp, $\pm$22pp), multi-seed confirmation would strengthen confidence. RULER multi-needle evaluation (8 trials) has insufficient statistical power for reliable comparison.
- **Downstream task coverage.** The passkey-retrieval mix demonstrates extrapolation benefit on a specific retrieval task. Standard long-context benchmarks (LongBench, SCROLLS) on instruction-tuned models remain future work.

7 Conclusion

We have recast RoPE frequency allocation as a variational inverse problem. A joint phase-collision–Fisher-fidelity functional, after broadband scale separation, yields a governing ODE whose exact solution (Theorem 1) is a cosh tether modulated by a Fisher pulse. The solution admits closed-form exact variational quantization (EVQ) with a single physical parameter τ that degrades exactly to geometric RoPE as $\tau \to 0$ (Theorem 2), proving that standard geometric RoPE is strictly suboptimal for any finite context (Corollary 1). The optimal $\tau^* = d_{\text{head}}/\sqrt{L}$ is derived from variational principles and validated across five context lengths.

Empirically, from-scratch training at four scales (50M–750M) on two datasets consistently confirms the theory's predictions. The most striking finding emerges at 750M scale: while both geometric and EVQ methods exhibit comparable OOD perplexity degradation (the waterbed effect), their positional retrieval trajectories fundamentally diverge—geometric RoPE *regresses* after 50% training while EVQ improves monotonically, reaching a +20pp gap. This *retrieval divergence* reveals that geometric RoPE is not merely statically suboptimal but dynamically degenerative: continued training actively damages its long-range capability. A passkey-mix experiment provides complementary evidence: EVQ doubles 4K extrapolation accuracy (42%→82%), and an antisymmetric data-scaling

phenomenon (Geo -22pp / EVQ +22pp when doubling mix ratio) demonstrates that frequency allocation quality, not data quantity, is the bottleneck for length generalization.

The entire method replaces one line of `inv_freq` initialization with zero hyperparameters. The degradation safety guarantee (Theorem 2: worst case = geometric) eliminates the risk of catastrophic failure. Future work includes warp-boundary (r^*) validation, layer-wise frequency allocation, task-conditional priors, and validation at billion-parameter scale. Broader-impact considerations are provided after References.

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Broader Impact

This work is a methodological framework for long-context positional design. Potential positive impact is improved stability diagnostics for context extension, which can reduce trial-and-error retraining cost when adapting models to longer windows. Potential negative impact is that longer-context capability can also strengthen misuse scenarios (e.g., scalable synthesis of harmful content and broader retrieval of sensitive information). Our contribution does not release new high-risk models or new datasets; nevertheless, we recommend standard deployment controls for long-context systems, including access restriction, content safeguards, and monitoring for privacy-sensitive prompts.

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A Proofs of Theorems 1 and 2

Proof of Theorem 1 (exact solution of the governing ODE). The joint functional after scale separation is

$$\mathcal{J}[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\,d\phi + \frac{\beta}{2}\int_0^1\int_0^1 \rho(\phi_1)\rho(\phi_2)\min(\phi_1,\phi_2)\,d\phi_1\,d\phi_2 - \mu_F\int_0^1 \rho(\phi)b^{-2\phi}\,d\phi + \lambda\left(\int_0^1 \rho(\phi)\,d\phi - 1\right). \tag{14}$$

Setting the first variation to zero yields

$$\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi - \mu_F b^{-2\phi} + \lambda = 0. \tag{15}$$

Differentiating once in ϕ, $\alpha\rho'(\phi) + \beta\int_\phi^1 \rho(\psi)\,d\psi - \mu_F(-2\ln b)b^{-2\phi} = 0$, which gives the boundary condition $\rho'(1) = -\frac{2\mu_F \ln b}{\alpha}b^{-2}$. Differentiating again yields the non-homogeneous ODE

$$\alpha\rho''(\phi) - \beta\rho(\phi) = -\mu_F(2\ln b)^2 b^{-2\phi}. \tag{16}$$

With $\tau = \sqrt{\beta/\alpha}$ and $\gamma_F = \mu_F(2\ln b)^2/\alpha$, this becomes $\rho'' - \tau^2\rho = \gamma_F b^{-2\phi}$. The homogeneous part has solutions $\cosh(\tau\phi), \sinh(\tau\phi)$. A particular solution of the form $Pb^{-2\phi}$ gives $P = \gamma_F/(4(\ln b)^2 - \tau^2)$, valid when $\tau \neq 2\ln b$. Hence the general solution is $\rho^*(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi) + \frac{\gamma_F}{4(\ln b)^2-\tau^2}b^{-2\phi}$, with C_1, C_2 determined by $\rho'(1)$ and $\int_0^1 \rho\,d\phi = 1$. □

Proof of Theorem 2 (asymptotic degradation of EVQ). The EVQ warp is $\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}\big((1-u_k)\sinh\tau\big)$ with $u_k = (k+0.5)/N$. As $\tau \to 0$, $\sinh\tau \to \tau + \mathcal{O}(\tau^3)$ and $\operatorname{arcsinh}(x) \to x + \mathcal{O}(x^3)$ for small x. Hence

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}\big((1-u_k)\tau + \mathcal{O}(\tau^3)\big) = 1 - \frac{(1-u_k)\tau + \mathcal{O}(\tau^3)}{\tau} = u_k + \mathcal{O}(\tau^2). \quad (17)$$

But $u_k = (k+0.5)/N$ is precisely the midpoint quantisation of the uniform CDF on $[0,1]$, which corresponds to geometric (log-linear) RoPE spacing. □

B Proof of Proposition 1 (uniform prior)

We prove that a uniform distance prior yields asymptotically constant diagonal energy.

Setup. Let $D(\Delta) = \frac{1}{L}\mathbb{1}_{\Delta\in[0,L]}$. Then

$$\widehat{D}_b(\xi) = \int_0^L \frac{1}{L}\cos(\xi\Delta)\,d\Delta = \frac{\sin(\xi L)}{\xi L} = \operatorname{sinc}(\xi L). \quad (18)$$

Hence $E_{\text{diag}}(\phi) = \frac{1}{2}\big(1 + \operatorname{sinc}(2b^{-\phi}L)\big)$.

Uniform decay as $L \to \infty$. Fix b and $\phi \in [0,1]$. Then $\xi(\phi) = 2b^{-\phi} \geq 2/b > 0$. As $L \to \infty$, $\operatorname{sinc}(\xi(\phi)L) \to 0$ and $|\operatorname{sinc}(\xi(\phi)L)| \leq 1/(\xi(\phi)L)$. In particular, for any $\phi \in [0,1]$,

$$\left|E_{\text{diag}}(\phi) - \frac{1}{2}\right| = \frac{1}{2}|\operatorname{sinc}(\xi(\phi)L)| \leq \frac{1}{2\xi(\phi)L} \leq \frac{b}{4L} \to 0, \quad (19)$$

which implies uniform convergence $E_{\text{diag}}(\phi) \to 1/2$.

Conclusion. By Lemma 1, $\rho^\star(\phi) \propto 1/E_{\text{diag}}(\phi)$ converges to a constant density, corresponding to geometric RoPE allocation. □

C Proof of Proposition 2 (power-law prior, corrected Ci asymptotics)

This appendix corrects a common algebraic pitfall: using $\operatorname{Ci}(z) \approx \gamma + \ln z$ outside its validity regime. The expansion $\operatorname{Ci}(z) = \gamma + \ln z + \mathcal{O}(z^2)$ holds for $z \ll 1$; for $z \gg 1$, $\operatorname{Ci}(z)$ decays oscillatory as $\operatorname{Ci}(z) = \frac{\sin z}{z} + \mathcal{O}(1/z^2)$.

Power-law prior and diagonal energy. Let $D(\Delta) = \frac{1}{\Delta\ln L}\mathbb{1}_{\Delta\in[1,L]}$. Then

$$\widehat{D}_b(\xi) = \frac{1}{\ln L}\int_1^L \frac{\cos(\xi\Delta)}{\Delta}\,d\Delta = \frac{1}{\ln L}\int_\xi^{\xi L}\frac{\cos u}{u}\,du = \frac{\operatorname{Ci}(\xi L) - \operatorname{Ci}(\xi)}{\ln L}. \quad (20)$$

Hence

$$E_{\text{diag}}(\phi) = \frac{1}{2}\left(1 + \frac{\operatorname{Ci}(2b^{-\phi}L) - \operatorname{Ci}(2b^{-\phi})}{\ln L}\right). \quad (21)$$

Bulk asymptotics and sign (fixing the algebraic error). Let $\xi = 2b^{-\phi}$. For ξ (bulk region, ϕ not too close to 0 so that $\xi \ll 1$),

$$\operatorname{Ci}(\xi) \approx \gamma + \ln\xi = \gamma + \ln 2 - \phi\ln b. \quad (22)$$

For $\xi L = 2b^{-\phi}L$ (when $L \gg b$, $\xi L \gg 1$),

$$\operatorname{Ci}(\xi L) \approx \frac{\sin(\xi L)}{\xi L} \approx 0 \qquad \text{(oscillatory decay)}. \quad (23)$$

Therefore,

$$\widehat{D}_b(2b^{-\phi}) \approx \frac{0 - (\gamma + \ln 2 - \phi\ln b)}{\ln L} = \frac{\phi\ln b - \gamma - \ln 2}{\ln L}, \quad (24)$$

and

$$E_{\text{diag}}(\phi) \approx \frac{1}{2}\left(1 + \frac{\phi\ln b - \gamma - \ln 2}{\ln L}\right), \quad (25)$$

which is strictly increasing in ϕ. Therefore $\rho^\star(\phi) \propto 1/E_{\mathrm{diag}}(\phi)$ is strictly decreasing: a high-frequency bias.

Amplitude remains bounded. Using the affine bulk form, one obtains an end-to-end ratio

$$\frac{\rho^\star(0)}{\rho^\star(1)} = \frac{E_{\mathrm{diag}}(1)}{E_{\mathrm{diag}}(0)} \approx \frac{\ln L + \ln b - \gamma - \ln 2}{\ln L - \gamma - \ln 2} = 1 + \frac{\ln b}{\ln L - \gamma - \ln 2}, \tag{26}$$

which is $1 + \mathcal{O}(\ln b/\ln L)$ when $\ln L \gg \ln b$. This expression is a bulk proxy: at $\phi = 0$ (so $\xi = 2$), one should use the exact constant $\mathrm{Ci}(2)$ instead of $\gamma + \ln 2$. The boundary correction remains finite and does not alter the bounded-amplitude conclusion.

Takeaway. The incorrect step to avoid is replacing both $\mathrm{Ci}(\xi)$ and $\mathrm{Ci}(\xi L)$ by $\gamma + \ln(\cdot)$, which would cancel to $\ln L$ and erase ϕ-dependence. The correct regime split is: $\mathrm{Ci}(\xi) \approx \gamma + \ln \xi$ for $\xi \ll 1$ but $\mathrm{Ci}(\xi L) \approx 0$ for $\xi L \gg 1$. □

D Proof of Proposition 3 (resonance condition and deep local minima)

Let $D(\Delta) = \lambda\delta(\Delta - s) + (1-\lambda)\delta(\Delta - \ell)$. Then

$$E_{\mathrm{diag}}(\phi) = \frac{1}{2}\Big(1 + \lambda\cos(2b^{-\phi}s) + (1-\lambda)\cos(2b^{-\phi}\ell)\Big). \tag{27}$$

Resonance bound. Assume there exist $\phi_\star \in [0,1]$, integers m, n, and nonnegative errors $\varepsilon_s, \varepsilon_\ell$ such that

$$\big|2b^{-\phi_\star}s - (2m+1)\pi\big| \le \varepsilon_s, \qquad \big|2b^{-\phi_\star}\ell - (2n+1)\pi\big| \le \varepsilon_\ell. \tag{28}$$

Write

$$2b^{-\phi_\star}s = (2m+1)\pi + \delta_s, \quad 2b^{-\phi_\star}\ell = (2n+1)\pi + \delta_\ell, \quad |\delta_s| \le \varepsilon_s,\ |\delta_\ell| \le \varepsilon_\ell. \tag{29}$$

Then

$$\cos(2b^{-\phi_\star}s) = -\cos\delta_s, \qquad \cos(2b^{-\phi_\star}\ell) = -\cos\delta_\ell, \tag{30}$$

and therefore

$$E_{\mathrm{diag}}(\phi_\star) = \frac{1}{2}\Big(1 - \lambda\cos\delta_s - (1-\lambda)\cos\delta_\ell\Big) \le \frac{1}{2}\Big(1 - \lambda\cos\varepsilon_s - (1-\lambda)\cos\varepsilon_\ell\Big). \tag{31}$$

Using $\cos x \ge 1 - x^2/2$ for small x gives

$$E_{\mathrm{diag}}(\phi_\star) \le \frac{\lambda\varepsilon_s^2 + (1-\lambda)\varepsilon_\ell^2}{4} = \mathcal{O}(\varepsilon_s^2 + \varepsilon_\ell^2). \tag{32}$$

Hence near-resonant alignments create arbitrarily deep diagonal minima.

Consequence for the inverse law. By Lemma 1, $\rho^\star(\phi) \propto 1/E_{\mathrm{diag}}(\phi)$ spikes near such minima, which is brittle and can diverge from the optimizer of the full functional where off-diagonal coupling smooths the landscape. □ For fixed b, larger long-range distances (s, ℓ) increase phase oscillation counts over $\phi \in [0,1]$, heuristically increasing the chance of near-resonant points.

E Diagonal approximation and residual analysis (summary)

Expanding the kernel,

$$K(\phi_1, \phi_2) = \int D(\Delta)\cos(\omega(\phi_1)\Delta)\cos(\omega(\phi_2)\Delta)\, d\Delta. \tag{33}$$

Using $\cos a \cos b = \frac{1}{2}(\cos(a-b) + \cos(a+b))$ shows that $K(\phi,\phi)$ contains a constant $1/2$ plus an oscillatory term involving $\widehat{D}_b(\omega(\phi))$, yielding the diagonal surrogate. Off-diagonal terms involve $\widehat{D}_b(\omega(\phi_1) \pm \omega(\phi_2))$ and are small when D is sufficiently smooth and broadband. A representative finite-base audit at $b = 10^4$ reports an $\mathcal{O}(1/\ln b)$ residual of about 11%.

F Structural stability beyond the diagonal

F.1 Brownian covariance

In the broadband limit $b \to \infty$ under a power-law prior, the off-diagonal kernel converges to

$$K_{\text{off}}(\phi_1, \phi_2) \to \frac{1}{2} \min(\phi_1, \phi_2), \tag{34}$$

the covariance of Brownian motion. We use the regularized functional

$$\mathcal{J}[\rho] = \frac{1}{2} \int_0^1 \rho(\phi)^2 \, d\phi + \frac{1}{2} \int_0^1 \int_0^1 \rho(\phi_1)\rho(\phi_2) \min(\phi_1, \phi_2) \, d\phi_1 d\phi_2 + \mu \left(\int_0^1 \rho(\phi) \, d\phi - 1 \right). \tag{35}$$

The first variation yields

$$\rho(\phi) + \int_0^1 \rho(\psi) \min(\phi, \psi) \, d\psi + \mu = 0. \tag{36}$$

Differentiating once gives

$$\rho'(\phi) + \int_\phi^1 \rho(\psi) \, d\psi = 0, \tag{37}$$

and differentiating again gives

$$\rho''(\phi) - \rho(\phi) = 0, \qquad \rho'(1) = 0. \tag{38}$$

Hence $\rho \propto \cosh(1 - \phi)$. If one keeps only the pure off-diagonal term in (35), the resulting equation is different ($\rho'' = 0$), so the diagonal regularizer is essential to recover the observed bounded profile.

F.1.1 Uniform validity

At $\phi = 0$ directly evaluate the exact off-diagonal kernel:

$$K_{\text{off}}(0, \psi) = -\frac{1}{4 \ln b} \Big[\text{Ci}(1 - \omega_2) + \text{Ci}(1 + \omega_2) \Big], \tag{39}$$

where $\omega_2 = \omega(\psi) = b^{-\psi} \in (0, 1)$. Using a symmetric Taylor expansion around argument 1, $\text{Ci}(1 \pm \omega_2) = \text{Ci}(1) \pm \text{Ci}'(1)\omega_2 + \mathcal{O}(\omega_2^2)$, the first-order terms cancel, yielding the pointwise boundary correction

$$\rho_{\text{exact}}(0) - \rho_{\text{out}}(0) = \frac{\text{Ci}(1)}{2 \ln b} \approx 0.018 \qquad (b = 10{,}000). \tag{40}$$

Since $\cosh(1) \approx 1.54$, the discrepancy is only $\sim 1.2\%$. There is no boundary singularity.

F.2 Non-commutativity

Under a uniform prior, the physically correct limit is $L \to \infty$ with finite d; reversing the order ($d \to \infty$ first) introduces a Jacobian distortion that changes the inferred density. This explains why some continuous approximations can contradict discrete behavior; see the discussion in Remark 1.

F.3 Fredholm correction

At finite base, the full optimization yields a Fredholm integral equation. We expand

$$\rho = \rho_0 + \varepsilon \rho_1, \qquad \varepsilon = 1/\ln b, \qquad \rho_0 = \cosh(1 - \phi), \tag{41}$$

and obtain the first-order correction

$$\rho_1(\phi) = -\int_0^1 \min(\phi, \psi) \cosh(1 - \psi) \, d\psi. \tag{42}$$

This admits the closed form

$$\rho_1(\phi) = \cosh(1 - \phi) - \cosh(1), \tag{43}$$

which is strictly negative for all $\phi > 0$. Intuitively, it acts as a uniform flattening penalty away from the high-frequency endpoint. The corrected solution is

$$\rho(\phi) \approx (1 + \varepsilon) \cosh(1 - \phi) - \varepsilon \cosh(1). \tag{44}$$

Reconciling remark. The b-independence of the leading $\cosh(1 - \phi)$ profile reflects the $b \to \infty$ broadband limit, where diagonal variation is suppressed. The Fredholm correction reintroduces finite-base dependence through $\varepsilon = 1/\ln b$. Thus Theorem 2 (diagonal, finite b) and the cosh solution (full, broadband limit) describe complementary regimes rather than conflicting predictions.

F.4 Waterbed inequality

Define $\mathcal{I}(\phi) = c\rho(\phi)b^{-2\phi}$ and note the proxy relation $E(\phi) \geq 1/\mathcal{I}(\phi)$. By Jensen's inequality,

$$\int_0^1 \ln E(\phi)\, d\phi \geq \ln b - \ln c. \tag{45}$$

This is an information-theoretic scaling bound: it formalizes an integrated trade-off between phase-collision suppression and local resolution, without claiming a pointwise tight operational bound.

G Additional non-asymptotic and discretization guarantees

Proposition G.1 (Broadband decomposition and structural invariance). *Let* $D(\Delta) = \frac{1}{\Delta \ln L}\mathbb{1}_{\Delta\in[1,L]}$. *In the broadband limit, the full kernel decomposes into a diagonal singular ridge plus a smooth off-diagonal Brownian covariance. Modeling the diagonal ridge with a Tikhonov weight* $\alpha > 0$*, consider*

$$\mathcal{J}_\alpha[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\, d\phi + \frac{1}{2}\int_0^1\int_0^1 \rho(\phi_1)\rho(\phi_2)\min(\phi_1,\phi_2)\, d\phi_1 d\phi_2 + \mu\left(\int_0^1 \rho(\phi)\, d\phi - 1\right). \tag{46}$$

Its optimizer is unique (up to normalization) and satisfies

$$\rho^\star_\alpha(\phi) \propto \cosh\left(\frac{1-\phi}{\sqrt{\alpha}}\right), \tag{47}$$

hence is strictly decreasing and strictly convex on $[0, 1)$.

Proof. For $\phi_1 \neq \phi_2$ and (without loss of generality) $\phi_1 > \phi_2$, we have $\omega_1 \ll \omega_2$ and therefore $\omega_1 \pm \omega_2 = \pm\omega_2 + o(1)$. Using $\widehat{D}_b(\xi) = \frac{\mathrm{Ci}(\xi L)-\mathrm{Ci}(\xi)}{\ln L}$ with $\mathrm{Ci}(\xi L) \approx 0$ and $\mathrm{Ci}(\xi) \approx \gamma + \ln \xi$ in the broadband asymptotic regime yields an off-diagonal term affine in the smaller log-frequency coordinate, i.e., proportional to $\min(\phi_1, \phi_2)$. At $\phi_1 = \phi_2$, $\widehat{D}_b(0) = 1$ creates a diagonal ridge contribution; in the continuous model we keep that contribution via the local L_2 term with weight α.

The first variation of (46) gives

$$\alpha\rho(\phi) + \int_0^1 \rho(\psi)\min(\phi,\psi)\, d\psi + \mu = 0. \tag{48}$$

Differentiating once,

$$\alpha\rho'(\phi) + \int_\phi^1 \rho(\psi)\, d\psi = 0, \tag{49}$$

which implies $\rho'(1) = 0$. Differentiating again yields

$$\alpha\rho''(\phi) - \rho(\phi) = 0, \qquad \rho'(1) = 0. \tag{50}$$

Thus $\rho(\phi) = A\cosh((1-\phi)/\sqrt{\alpha})$, and normalization fixes A. Since $\sinh(x) > 0$ and $\cosh(x) > 0$ for $x > 0$, we have $\rho'_\alpha(\phi) < 0$ and $\rho''_\alpha(\phi) > 0$ on $[0, 1)$. Strict convexity of $\mathcal{J}_\alpha$ for $\alpha > 0$ gives uniqueness. □ Physically, α controls the relative strength of the diagonal ridge; for finite discrete heads, this term reflects the non-vanishing diagonal mass contribution (order $1/N$). In the formal limit $\alpha \to 0^+$, the model degenerates to the pure Brownian regime with linear stationarity, while any $\alpha > 0$ preserves the observed monotone-convex high-frequency bias.

Proposition G.2 (Finite-(b, L) residual bound for the affine surrogate). *Let*

$$\widetilde{E}_{\mathrm{diag}}(\phi) = \frac{1}{2}\left(1 + \frac{\phi\ln b - \gamma - \ln 2}{\ln L}\right), \tag{51}$$

and define $\Delta E(\phi) = |E_{\mathrm{diag}}(\phi) - \widetilde{E}_{\mathrm{diag}}(\phi)|$. *For finite* b, L*:*

$$\Delta E(\phi) \leq \frac{1}{2\ln L}\left(b^{-2\phi} + \frac{b^\phi}{L}\right). \tag{52}$$

Proof. From (21),

$$E_{\mathrm{diag}}(\phi) = \frac{1}{2}\left(1 + \frac{\mathrm{Ci}(2b^{-\phi}L) - \mathrm{Ci}(2b^{-\phi})}{\ln L}\right). \tag{53}$$

For $x > 0$, using $\mathrm{Ci}(x) = \gamma + \ln x - \int_0^x \frac{1-\cos t}{t}\,dt$ and $1 - \cos t \le t^2/2$,

$$|\mathrm{Ci}(x) - (\gamma + \ln x)| \le \int_0^x \frac{t}{2}\,dt = \frac{x^2}{4}. \tag{54}$$

At $x = 2b^{-\phi}$, this gives a near-field error $\le b^{-2\phi}$. For $X > 0$, integration by parts implies

$$|\,\mathrm{Ci}(X)| = \left|\frac{\sin X}{X} - \int_X^\infty \frac{\sin t}{t^2}\,dt\right| \le \frac{2}{X}, \tag{55}$$

so for $X = 2b^{-\phi}L$ the far-field error is $\le b^{\phi}/L$. Combining by triangle inequality and dividing by $2\ln L$ yields (52). □ For the two finite-base settings used in Section 4.3 ($L = 16384$ with $b = 10^4$ and $b = 5 \times 10^5$), this bound is consistent with the observed high-R^2_{mid} calibration and small mid-band mean absolute residuals.

Proposition G.3 (Regularized inverse law and concentration collapse). *Assume a resonant trap at $\phi_\star$ with local behavior $E_{\mathrm{diag}}(\phi) \approx c(\phi - \phi_\star)^2$, $c > 0$. Define*

$$\rho^\star_\epsilon(\phi) = \frac{1}{Z_\epsilon}\frac{1}{E_{\mathrm{diag}}(\phi) + \epsilon}, \qquad \epsilon > 0. \tag{56}$$

Then $\|\rho^\star_\epsilon\|_\infty = \mathcal{O}(1/\sqrt{\epsilon})$. As $\epsilon \to 0^+$, $\rho^\star_\epsilon$ converges weakly to $\delta(\phi - \phi_\star)$.

Proof. Near $\phi_\star$ we approximate

$$Z_\epsilon \approx \int_{-\infty}^{\infty} \frac{d\phi}{c(\phi - \phi_\star)^2 + \epsilon} = \frac{\pi}{\sqrt{c\epsilon}}. \tag{57}$$

Hence

$$\rho^\star_\epsilon(\phi_\star) = \frac{1}{\epsilon Z_\epsilon} = \frac{\sqrt{c}}{\pi\sqrt{\epsilon}} = \mathcal{O}\left(\frac{1}{\sqrt{\epsilon}}\right). \tag{58}$$

For any continuous test function f, the normalized Cauchy kernel in (56) satisfies

$$\lim_{\epsilon \to 0^+} \int_0^1 f(\phi)\rho^\star_\epsilon(\phi)\,d\phi = f(\phi_\star), \tag{59}$$

which is weak convergence to $\delta(\phi - \phi_\star)$. □ In sparse discrete settings (e.g., $d = 128$ and $N = 64$ frequencies), this collapse appears as density concentrating on very few bins, effectively starving the remaining frequency coordinates.

Proposition G.4 (Pointwise off-diagonal decay for distinct frequencies). *Under $D(\Delta) = \frac{1}{L}\mathbb{1}_{\Delta \in [0,L]}$, let*

$$K_{\mathrm{off}}(\phi_1, \phi_2) = \frac{1}{2}\left[\mathrm{sinc}((\omega_1 - \omega_2)L) + \mathrm{sinc}((\omega_1 + \omega_2)L)\right], \quad \omega_i = b^{-\phi_i}. \tag{60}$$

For any pair with $\phi_1 \neq \phi_2$, $K_{\mathrm{off}}(\phi_1, \phi_2) = \mathcal{O}(1/L)$ as $L \to \infty$. Moreover, on any separated set $\{|\omega_1 - \omega_2| \ge \delta > 0\}$, the bound is uniform.

Proof. Using $|\mathrm{sinc}(x)| \le 1/|x|$,

$$|K_{\mathrm{off}}(\phi_1, \phi_2)| \le \frac{1}{2}\left(\frac{1}{|\omega_1 - \omega_2|L} + \frac{1}{(\omega_1 + \omega_2)L}\right) = \mathcal{O}\left(\frac{1}{L}\right), \tag{61}$$

for fixed (ϕ_1, ϕ_2) with $\omega_1 \neq \omega_2$. If $|\omega_1 - \omega_2| \ge \delta$, then

$$|K_{\mathrm{off}}(\phi_1, \phi_2)| \le \frac{1}{2L}\left(\frac{1}{\delta} + \frac{1}{\omega_1 + \omega_2}\right) \le \frac{C_\delta}{L}, \tag{62}$$

which is uniform on the separated set. □ Under the power-law prior, an analogous asymptotic argument gives logarithmic attenuation with L for corresponding off-diagonal terms.

Proposition G.5 (Quantization error and anchoring necessity). *Let* $\phi_k = F^{-1}\big(\frac{k+0.5}{N}\big)$ *with* $F(\phi) = \int_0^\phi \rho(t)\,dt$*, and define* $g(u) = E_{\text{diag}}(F^{-1}(u))$ *on* $u \in [0,1]$*. Then midpoint quantization gives*

$$\left| \frac{1}{N} \sum_{k=0}^{N-1} g\left(\frac{k+0.5}{N}\right) - \int_0^1 g(u)\,du \right| \le \frac{1}{24N^2} \sup_{u\in[0,1]} |g''(u)|. \tag{63}$$

Moreover,

$$g''(u) = \frac{E''_{\text{diag}}(\phi)}{\rho(\phi)^2} - \frac{E'_{\text{diag}}(\phi)\rho'(\phi)}{\rho(\phi)^3}, \tag{64}$$

hence

$$\mathcal{E}_{\text{quant}} \le \frac{1}{24N^2} \sup_{\phi\in[0,1]} \left| \frac{E''_{\text{diag}}(\phi)}{\rho(\phi)^2} - \frac{E'_{\text{diag}}(\phi)\rho'(\phi)}{\rho(\phi)^3} \right|. \tag{65}$$

If $\inf_\phi \rho(\phi) \to 0$*, the bound can blow up; enforcing* $\inf_\phi \rho(\phi) \ge c_0 > 0$ *through anchoring prevents this singular behavior.*

Proof. With $u = F(\phi)$ and $d\phi/du = 1/\rho(\phi)$, midpoint quadrature on g yields (63). By chain rule,

$$g'(u) = E'_{\text{diag}}(\phi)\frac{d\phi}{du} = \frac{E'_{\text{diag}}(\phi)}{\rho(\phi)}, \tag{66}$$

and differentiating again gives (64). Substituting into (63) yields (65). The $1/\rho^3$ term shows why near-zero density creates discretization instability, while a positive density floor prevents blow-up. □

H Experimental Hyperparameter Configurations

This appendix summarizes the reproducibility-critical settings used for all reported results.

Item	Configuration
Models	50M (3 seeds: 42/137/256), 125M (single seed), 350M (single seed)
Dataset	TinyStories [22]
RoPE base	$b = 500{,}000$
EVQ parameter	$\tau = 1.5$ (predicted: $64/\sqrt{2048} \approx 1.41$)
Train/eval lengths	Train at 2048; evaluate at 2K, 4K, 8K, 16K
Tokens	50M per run

H.1 From-scratch TinyStories runs (Table 2, top 3 rows).

Item	Configuration
Model	350M (hidden=1024, layers=24, heads=16, head_dim=64)
Dataset	FineWeb-Edu (streaming from HuggingFace)
Seeds	42, 137, 256
RoPE base	$b = 500{,}000$; $\tau = 1.5$
Tokens	50M per run
GPU	NVIDIA RTX 5090

H.2 350M FineWeb-Edu 3-seed (Tables 2–3).

Item	Configuration
Model	350M (same architecture as H.2)
Dataset	90% FineWeb-Edu + 10% passkey-retrieval
RoPE base	$b = 500{,}000$; $\tau = 1.5$
Train length	2048; evaluate retrieval at 2K, 4K, 8K; PPL at 2K, 8K, 16K
Seed	42

H.3 Passkey-retrieval mix (Table 5).

Item	Configuration
Model	750M (hidden=1536, layers=18, heads=24, head_dim=64, FFN=6144)
Dataset	FineWeb-Edu (streaming) + 3% passkey mix
Methods	Geometric vs Hybrid (τ=1.5, r=16)
Tokens	1B per method (15258 steps)
Checkpoints	25%, 50%, 75%, 100% of training
Seed	42
Eval: PPL	L $\in$ {1K, 2K, 4K, 8K, 16K}, 4 chunks per checkpoint, 6 chunks final
Eval: Passkey	L $\in$ {1K, 2K, 4K, 8K}, depth=0.5, 20 trials/checkpoint, 40 trials final
Eval: RULER	5 needles, L $\in$ {2K, 4K, 8K}, 8 trials at 75%+100%
GPU	RTX PRO 6000 96GB
Training time	16.4h total (491.8min Geo + 492.0min Hybrid)

H.4 750M training dynamics (Table 6, Figure 1).

Item	Configuration
Model	50M
Methods	Geometric, EVQ ($\tau = 1.5$), YaRN (progressive), PI
Budget control	Matched training budget and data pipeline
Evaluation lengths	2K, 4K, 8K, 12K, 16K

H.5 Competitor comparison (Table 4).

Item	Configuration
Model	50M
L_{train} values	128, 256, 512, 1024, 2048
τ sweep	[0.0, 0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0] per L
RoPE base	$b = 500{,}000$

H.6 τ^* scaling law sweep (Table 7).

Item	Configuration
Model	350M
RoPE base	$b = 10{,}000$
L_{train}	4096
τ values	0.0 (Geometric), 0.7, 1.1
Tokens	50M

H.7 Collision-block dead-zone validation (Table 8).

H.8 EVQ-cosh implementation. The complete EVQ schedule is:

```
def evq_cosh_inv_freq(d_head, L, base):
    tau = d_head / math.sqrt(L * math.log(base))
    K = d_head // 2
    u = torch.linspace(0.5/K, 1 - 0.5/K, K)
    phi = 1 - (1/tau) * torch.arcsinh(
        (1 - u) * math.sinh(tau))
    return base ** (-phi)
```

This replaces the standard `inv_freq = base ** (-torch.arange(0, d, 2) / d)` with zero additional hyperparameters.